Applied Cryptography

Lecture 7: Provable Security and a Security Proof for CTR Mode

Kien Tuong Truong (kitruong@ethz.ch)

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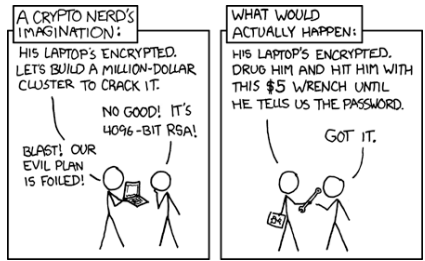
What Does *Secure* Mean?

At the end of the lecture we will prove that CTR mode is *secure*.
Later in the course, you will see protocols (e.g. TLS, Signal) that are also considered *secure*.

What do you think it means for a scheme/protocol to be secure?

What Does *Secure* Mean?

A possible answer: that there are no attacks against it.
Sounds nice, but turns out to be impossible to achieve.



Another answer: that *within a well-defined security model*, no adversary can "efficiently" attack the scheme/protocol.

If we say that "AES in CTR mode is (q, t, ε)-IND-CPA secure" we mean that:

- The AES-CTR *construction*...
- (which is a *symmetric encryption scheme*)...
- is secure in the IND-CPA *security model*...
- meaning that any adversary $\mathcal{A}$ playing in the IND-CPA game against AES-CTR...
- running in at most time t...
- making at most q queries...
- has advantage *at most* ε.

The IND-CPA game states what the adversary can and cannot do.
The value ε measures how successful the adversary is.
The values q and t define what "efficient" means.

What Does *Provable* Mean?

This is good, but how do we know that e.g. AES-CTR is actually (q, t, ε)-IND-CPA secure?

One could argue *heuristically* by trying to attack the scheme (and failing), but this is not a rigorous argument.

We will follow a formal *reductionist* approach.

Reductionist Approach

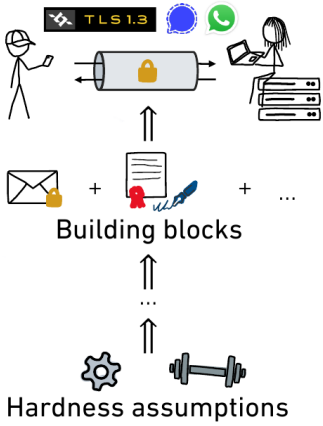


Image courtesy of Prof. Matilda Backendal

A protocol's security is *reduced* to the security of its components.

Its components' security is reduced to the security of their components, and so on.

In the end, we usually reduce to a *hardness assumption*, which we cannot prove.

Reductionist Approach (cont.)

Q: But why even do reductions, if in the end we have to *assume* something?

A: Because provable security forces us to explicitly state our assumptions. These assumptions are significantly easier to analyze (usually) and many of them are well-studied.

Reductionist Approach (cont.)

$$
\begin{array}{l}
\underline{\mathrm{CTR}[\mathrm{E}].\mathrm{Enc}(K, m)}: \\
\mathrm{ctr} \xleftarrow{\$} \{0,1\}^n \\
\textbf{for } i = 0 \textbf{ to } \ell - 1 \textbf{ do} \\
\quad r_i \leftarrow E_K(\mathrm{ctr} + i \bmod 2^n) \\
\quad c_i \leftarrow m_i \oplus r_i \\
\textbf{return } \mathrm{ctr} \| c_0 \| c_1 \| \cdots \| c_{\ell-1}
\end{array}
$$

We will prove that the security of CTR mode reduces to the security of the block cipher E!

Reductionist Approach (cont.)

Q: But why even do reductions, if we are only working in a model of reality?

> *"All models are wrong, but some are useful."*

A: Because *all* attacks must either *attack the assumptions directly* or exploit something *outside of the model*.

But I thought this was *Applied* Cryptography...

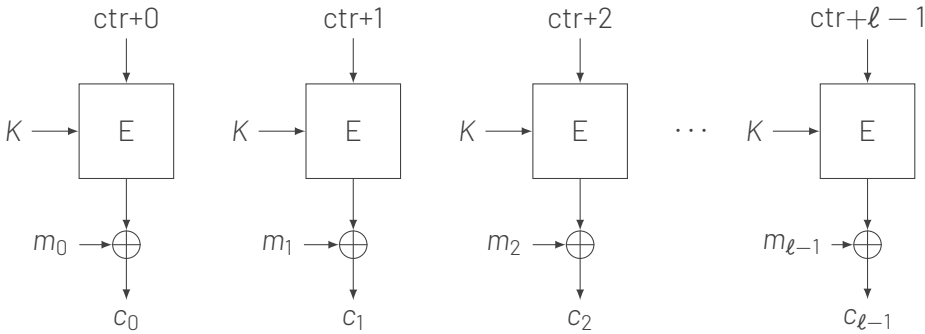
Provable security *is* applied cryptography!

Many modern systems – TLS, Signal, Whatsapp, iMessage, among others – have been analyzed using provable security techniques and reduced to well-understood assumptions.

The goal of applied cryptography is to give strong guarantees about the security of real-world systems. Provable security allows us to do this in a rigorous way.

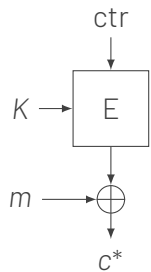
CTR Mode

Let E be a block cipher with block size n and key space $\{0,1\}^k$.
From previous lectures CTR[E] is:



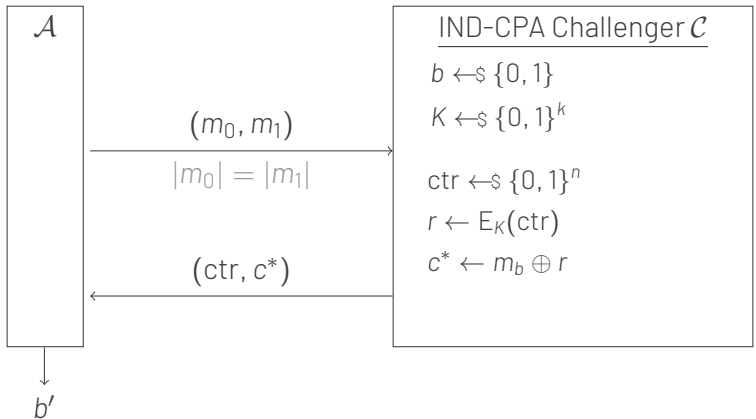
CTR Mode (Simplified)

Today, we will prove security for one-block messages $m \in \{0,1\}^n$ and random counters $\text{ctr} \leftarrow\!\!\$\ \{0,1\}^n$.



The ciphertext is $c = (\text{ctr}, c^*)$.

IND-CPA Security of CTR Mode



IND-CPA Advantage

The advantage of an adversary $\mathcal{A}$ in the IND-CPA game is:

$$\mathrm{Adv}^{\text{IND-CPA}}_{\text{CTR[E]}}(\mathcal{A}) = 2\left|\Pr[b' = b] - \frac{1}{2}\right|$$

CTR[E] is (q, t, ε)-IND-CPA secure if for all adversaries $\mathcal{A}$ running in *at most* time t, making *at most* q queries, the advantage is *at most* ε

We would like to relate this to the security of E.

Security of Block Ciphers

Consider CTR mode constructed using a block cipher E. Which assumption of *E* should we reduce to?

1. *E* is IND-CPA secure
2. *E* is a PRF
3. *E* is a PRP

PRP security is the natural notion for block ciphers.

Game Hopping Proofs

A *direct reduction* (recall Exercise 2) can be used to prove this, but it's tricky.

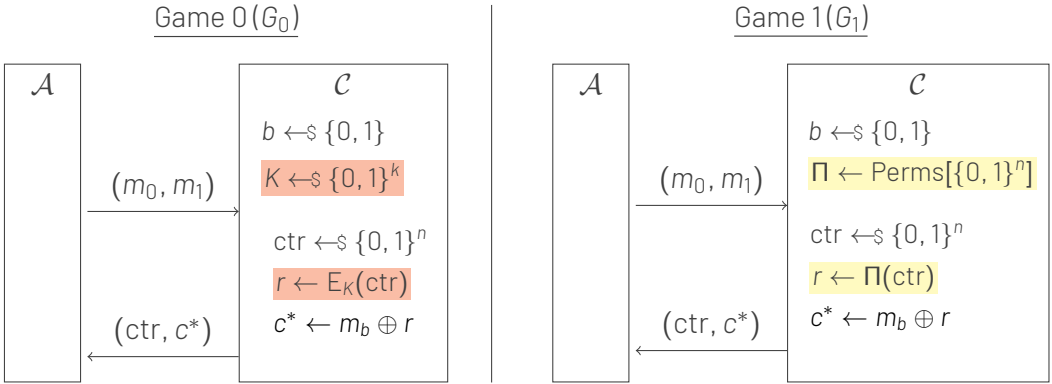
The *game hopping* framework is a way to structure proofs in a modular way, introduced in [KR96].

Some useful resources:

- [BR04] *"The Game-Playing Technique"* by Bellare and Rogaway.
- [Sho06] *"Sequences of Games: A Tool for Taming Complexity in Security Proofs"* by Shoup.
- Our HA Andrea Raguso has written a nice summary of the technique (see Moodle).

Intuition

Imagine the *same* adversary $\mathcal{A}$ in the two following games.



Intuition (cont.)

Which of the following is true?

1. $\mathcal{A}$'s advantage in $G_0 \gg \mathcal{A}$'s advantage in G_1
2. $\mathcal{A}$'s advantage in $G_0 \approx \mathcal{A}$'s advantage in G_1
3. $\mathcal{A}$'s advantage in $G_0 \ll \mathcal{A}$'s advantage in G_1

Why?

Hopping Between G_0 and G_1

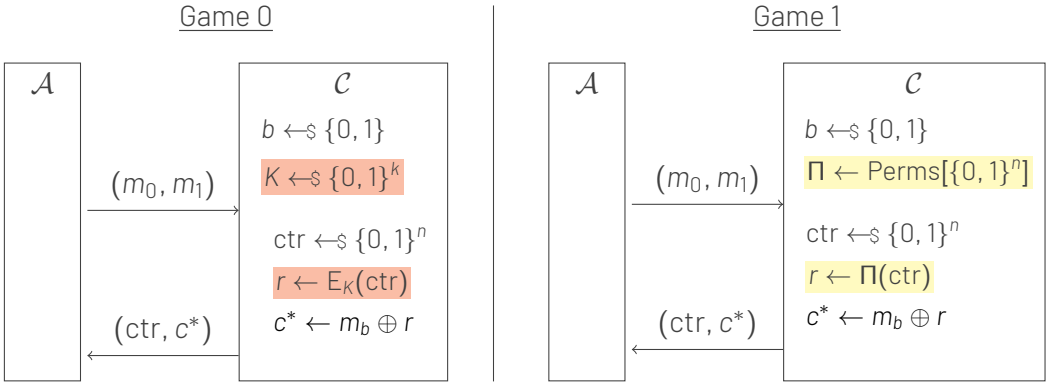
Claim

The advantage must be approximately the same. Otherwise, E is not PRP secure.

Proof Intuition

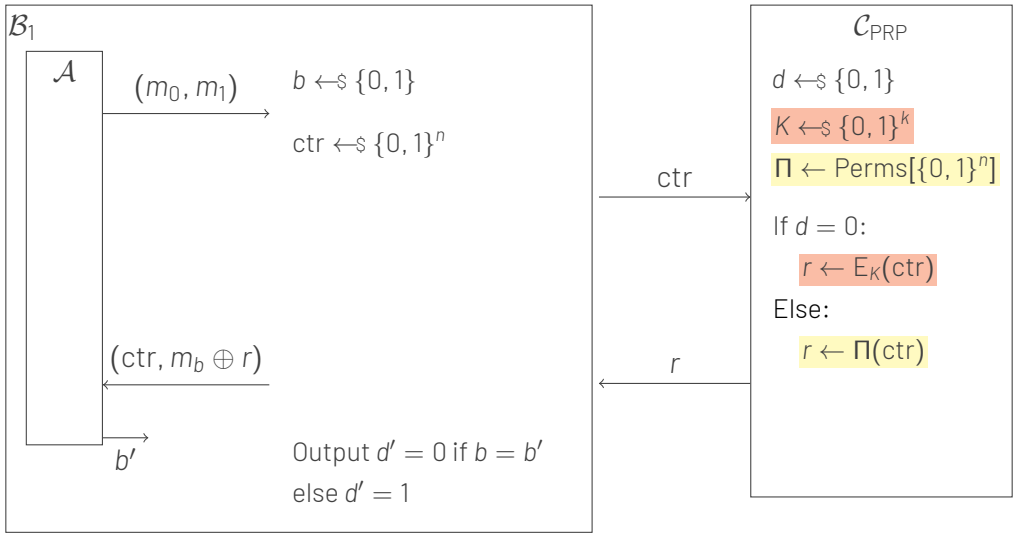
We will build a PRP adversary $\mathcal{B}$ which uses $\mathcal{A}$ as a subroutine and uses its challenger to simulate either G_0 or G_1 to $\mathcal{A}$. If $\mathcal{A}$ *behaves differently* between the two games, i.e. its advantage is much larger in G_0 than in G_1 (or viceversa), then $\mathcal{B}$ can exploit this to win its PRP game

Hopping Between G_0 and G_1 (cont.)



The *highlighted* parts can be "refactored" into a real-or-random PRP challenger.

Hopping Between G_0 and G_1 (cont.)



Bounding the Difference in Advantage

Let:

$$q_0 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_0]$$

$$q_1 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_1].$$

We want to show that $|q_0 - q_1|$ is small. We have:

$$q_0 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_0] = \Pr[d' = 0 | d = 0]$$

$$q_1 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_1] = \Pr[d' = 0 | d = 1].$$

Via advantage rewriting, we conclude that:

$$|q_0 - q_1| = |\Pr[d' = 0 | d = 0] - \Pr[d' = 0 | d = 1]| = \mathrm{Adv}^{\mathrm{PRP}}_{\mathrm{E}}(\mathcal{B}_1)$$

so if $|q_0 - q_1|$ was large, then E would not be a secure PRP.

Discussing the Bound

Important: the running time of $\mathcal{B}_1$ is approximately the same as the running time $t_{\mathcal{A}}$ of $\mathcal{A}$. If $\mathcal{A}$ makes q queries to its left-or-right oracle, then $\mathcal{B}_1$ makes q queries to its real-or-random oracle.

Main consequence: we can bound $|q_0 - q_1|$ by the *maximum* advantage of *any* PRP adversary $\mathcal{D}$ against E that runs in time $t_{\mathcal{A}}$ and makes q queries.

Why? $\mathcal{B}_1$ is a *specifically constructed* PRP adversary and can't have a better advantage than *the best* PRP adversary.

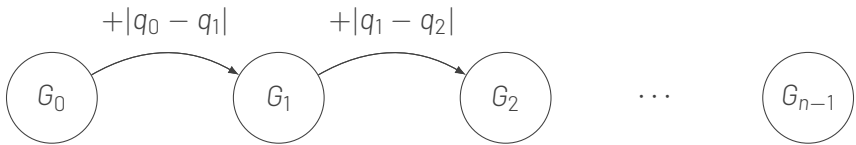
Game Hopping, in General

We moved from our original IND-CPA game G_0 to a "more ideal" game G_1.

Still, G_0 and G_1 are "close" in terms of the adversary's advantage.

Idea: extend this, and keep switching to new games (which are close to each other) until we reach a game where the advantage is known (e.g. 0)

Game Hopping, visualized



Let $q_i = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_i]$. Then, for a bit-guessing game, we have:

$$\left|q_0 - \frac{1}{2}\right| = \left|q_0 - q_1 + q_1 - q_2 + \ldots + q_{n-1} - \frac{1}{2}\right|$$

$$\leq |q_0 - q_1| + |q_1 - q_2| + \ldots + \left|q_{n-1} - \frac{1}{2}\right|$$

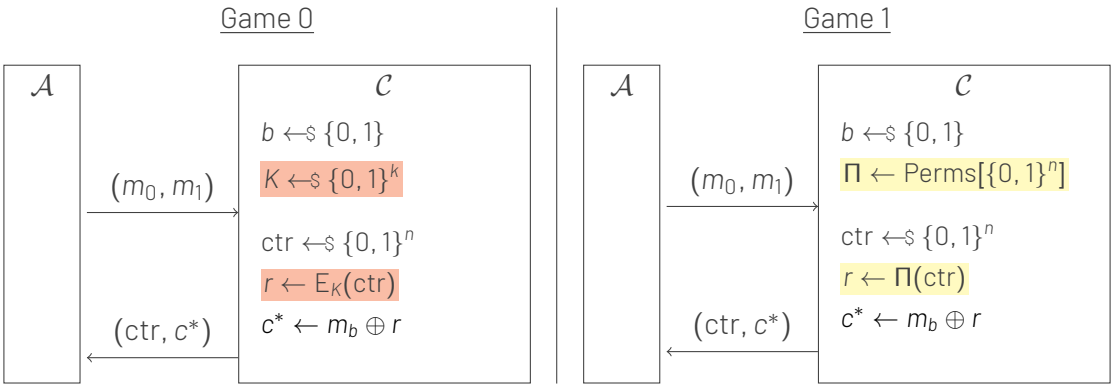
where $\left|q_{n-1} - \frac{1}{2}\right|$ is known.

Overview of the Proof

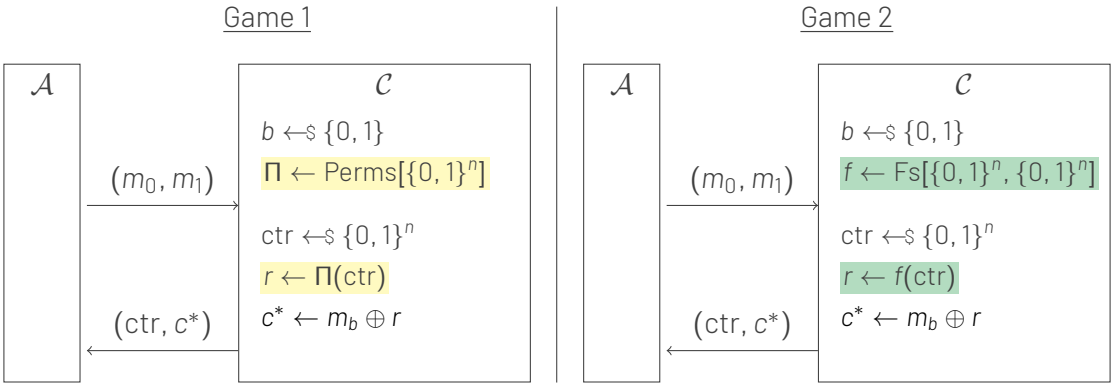
We are going to use a total of four games:

- Game 0: The original IND-CPA game
- Game 1: We replace E with a random permutation Π
- Game 2: We replace Π with a random *function f*
- Game 3: Instead of evaluating *f* on ctr, return a uniformly random value for *r*

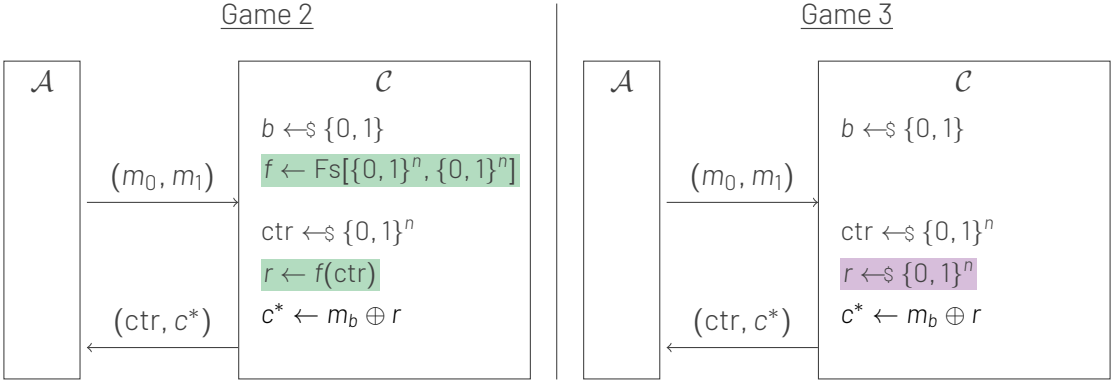
G_0 **VS** G_1


Game 0
$\mathcal{A}$
$\mathcal{C}$
$b \leftarrow\$ \{0,1\}$
$K \leftarrow\$ \{0,1\}^k$
(m_0, m_1)
$\text{ctr} \leftarrow\$ \{0,1\}^n$
$r \leftarrow \mathsf{E}_K(\text{ctr})$
(ctr, c^*)
$c^* \leftarrow m_b \oplus r$
Game 1
$\mathcal{A}$
$\mathcal{C}$
$b \leftarrow\$ \{0,1\}$
$\Pi \leftarrow \text{Perms}[\{0,1\}^n]$
(m_0, m_1)
$\text{ctr} \leftarrow\$ \{0,1\}^n$
$r \leftarrow \Pi(\text{ctr})$
(ctr, c^*)
$c^* \leftarrow m_b \oplus r$

G_1 VS G_2


Game 1
$\mathcal{A}$
$\mathcal{C}$
$b \leftarrow\$ \{0,1\}$
$\Pi \leftarrow \mathsf{Perms}[\{0,1\}^n]$
(m_0, m_1)
$\mathsf{ctr} \leftarrow\$ \{0,1\}^n$
$r \leftarrow \Pi(\mathsf{ctr})$
(ctr, c^*)
$c^* \leftarrow m_b \oplus r$
Game 2
$\mathcal{A}$
$\mathcal{C}$
$b \leftarrow\$ \{0,1\}$
$f \leftarrow \mathsf{Fs}[\{0,1\}^n, \{0,1\}^n]$
(m_0, m_1)
$\mathsf{ctr} \leftarrow\$ \{0,1\}^n$
$r \leftarrow f(\mathsf{ctr})$
(ctr, c^*)
$c^* \leftarrow m_b \oplus r$

G_2 **VS** G_3



What is the advantage of $\mathcal{A}$ in G_3? Zero! It's a one-time pad.

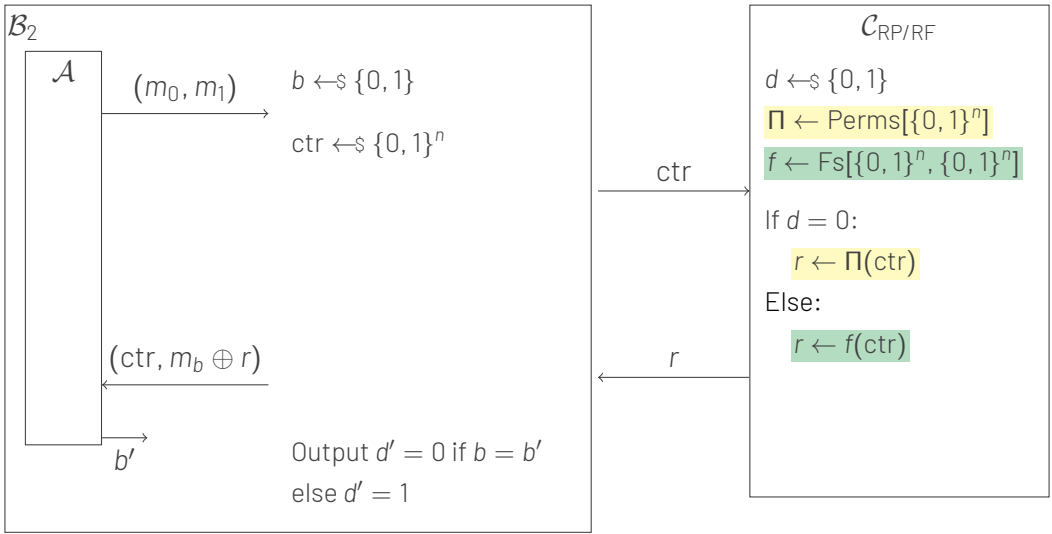
From G_0 to G_3

We have:

$$\frac{1}{2} \cdot \mathrm{Adv}^{\text{IND-CPA}}_{\text{CTR[E]}}(\mathcal{A}) = \left| q_0 - \frac{1}{2} \right| \leq |q_0 - q_1| + |q_1 - q_2| + |q_2 - q_3| + \left| q_3 - \frac{1}{2} \right|.$$

The last term is 0, and we already bounded the first term. We need to bound the other terms similarly.

Hopping Between G_1 and G_2



Bounding $|q_1 - q_2|$

As before:

$$q_1 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_1] = \Pr[d' = 0 | d = 0]$$
$$q_2 = \Pr[b' = b \mid \mathcal{A} \text{ plays in } G_2] = \Pr[d' = 0 | d = 1].$$

Via advantage rewriting, we have:

$$|q_1 - q_2| = |\Pr[d' = 0 | d = 0] - \Pr[d' = 0 | d = 1]| = \mathsf{Adv}^{\mathsf{RP/RF}}(\mathcal{B}_2)$$

That is, we bound the difference by the advantage of $\mathcal{B}_2$ of distinguishing between a random permutation and a random function.

Bounding $|q_1 - q_2|$ (cont.)

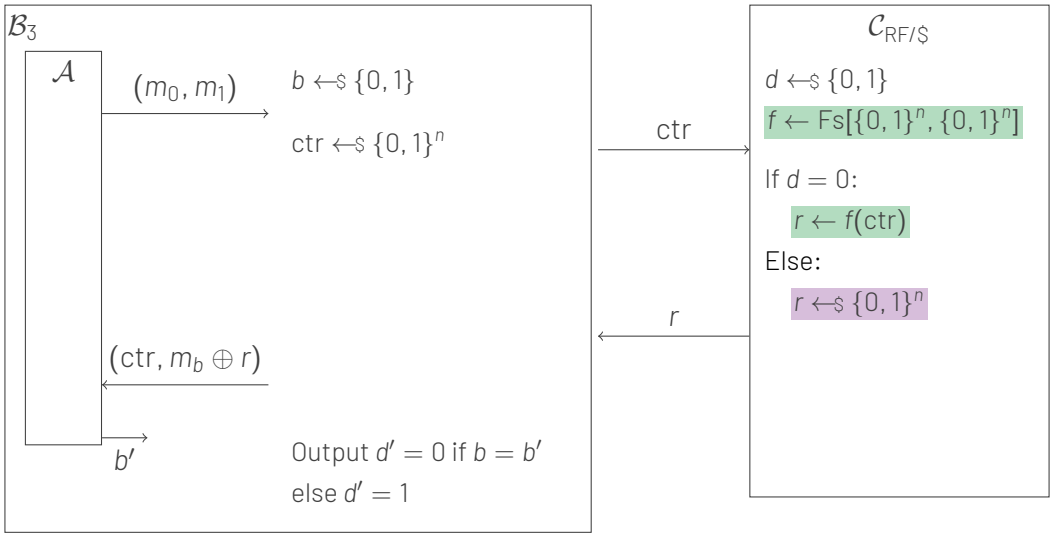
From the previous lectures, we know that for *any* algorithm $\mathcal{D}$ making q queries:

$$\mathrm{Adv}^{\mathrm{RP/RF}}(\mathcal{D}) \leq \frac{q^2}{2^{n+1}}$$

Hence:

$$|q_1 - q_2| = \mathrm{Adv}^{\mathrm{RP/RF}}(\mathcal{B}_2) \leq \frac{q^2}{2^{n+1}}$$

Hopping Between G_2 and G_3



Bounding $|q_2 - q_3|$

Question

What is the difference between using a *random function* and sampling a random value?

Answer

If ctr ever repeats, the random function's outputs will be consistent. If we instead sample random values, then we will likely get different values for *r* even when we use the same ctr.

Bounding $|q_2 - q_3|$ (cont.)

Let Z be the event that the randomly chosen values of ctr are *not all distinct*. If Z does not occur, then G_2 and G_3 are identical: the outputs of f on distinct inputs are *independent and uniformly random values*.

Let X_i be the event that $b' = b$ in game G_i (so $q_i = \Pr[X_i]$). Then:

$$\Pr[X_2 \wedge \neg Z] = \Pr[X_3 \wedge \neg Z].$$

Thus, by the difference lemma:

$$|q_2 - q_3| = |\Pr[X_2] - \Pr[X_3]| \leq \Pr[Z] \leq \frac{q^2}{2^{n+1}}.$$

Putting it all Together

To conclude:

$$\frac{1}{2} \cdot \mathrm{Adv}^{\text{IND-CPA}}_{\text{CTR[E]}}(\mathcal{A})$$

$$\leq |q_0 - q_1| + |q_1 - q_2| + |q_2 - q_3| + \left| q_3 - \frac{1}{2} \right|$$

$$\leq \mathrm{Adv}^{\text{PRP}}_{E}(\mathcal{B}_1) + \frac{q^2}{2^{n+1}} + \frac{q^2}{2^{n+1}} + 0 = \mathrm{Adv}^{\text{PRP}}_{E}(\mathcal{B}_1) + \frac{q^2}{2^n}.$$

Thus:

$$\mathrm{Adv}^{\text{IND-CPA}}_{\text{CTR[E]}}(\mathcal{A}) \leq 2 \cdot \mathrm{Adv}^{\text{PRP}}_{E}(\mathcal{B}_1) + \frac{q^2}{2^{n-1}}$$

for *any* adversary $\mathcal{A}$ making q queries.

Interpreting the Bound

We have proved the following theorem:

Theorem

If E is (q, t, ε)-PRP-secure, then CTR$[E]$ is $(q, t, 2\varepsilon + \frac{q^2}{2^{n-1}})$-IND-CPA-secure.

Note that the security slightly degrades. In fact, the bound becomes meaningless when q is large compared to $2^{n/2}$.
All known modes of operations have *data encryption limits*: there is a maximum amount of data that can be encrypted using the same key.

Can we get a Better Bound?

Not by much. CTR mode is broken if counters are re-used, so there is an attack that matches the bound. See exercises (week 4) for further details.

Homework

- **Action**: Try to extend the analysis to CTR mode with longer messages – main challenge is to bound collision probabilities for the counter values (see also exercise sheet 4).
- **Action**: Start exercise sheet 3 and prepare for lab 3.

Bibliography and Resources

[BR04] Mihir Bellare and Phillip Rogaway. "The Game-Playing Technique". In: (Dec. 2004). URL: https://cr.yp.to/bib/2004/bellare-games.pdf.

[KR96] Joe Kilian and Phillip Rogaway. "How to Protect DES Against Exhaustive Key Search". In: *Advances in Cryptology – CRYPTO'96*. Ed. by Neal Koblitz. Vol. 1109. Lecture Notes in Computer Science. Santa Barbara, CA, USA: Springer Berlin Heidelberg, Germany, Aug. 1996, pp. 252–267. DOI: 10.1007/3-540-68697-5_20.

[Sho06] Victor Shoup. "Sequences of Games: A Tool for Taming Complexity in Security Proofs". In: (Jan. 2006). URL: https://shoup.net/papers/games.pdf.